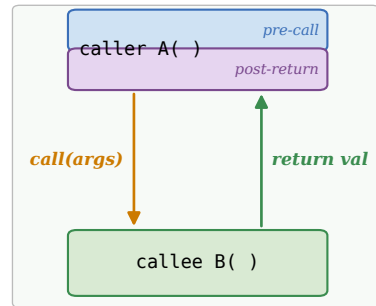
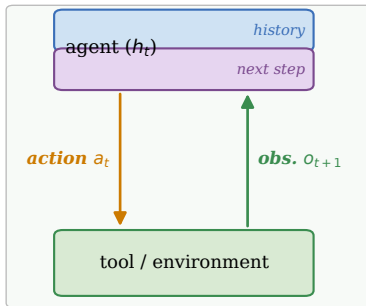


**Function call site**  $\sim$  **agent step** (structurally similar)  $\Rightarrow$  **use as self-supervised FIM mid-training prior**

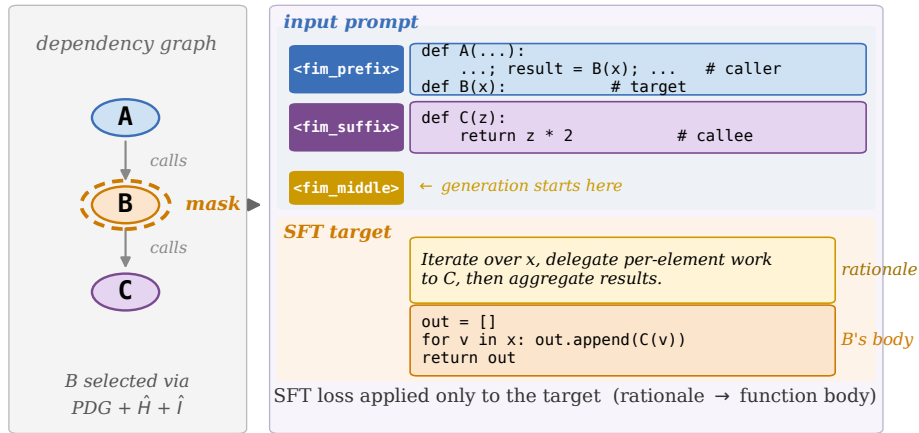
**Function call site**



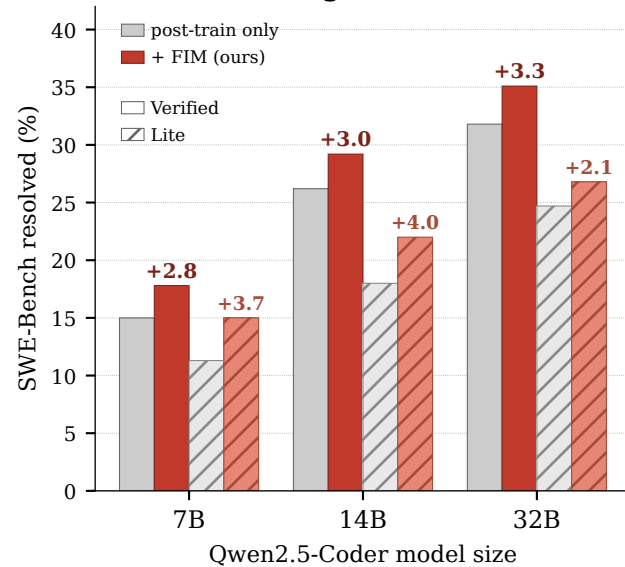
**Coding-agent step**



**Function-level FIM mid-training (SFT)**



**SWE-Bench gain across scales**



■ context     
 ■ call / action     
 ■ return / observation     
 ■ continuation  
 same 4-stage decomposition: context  $\rightarrow$  call  $\rightarrow$  return  $\rightarrow$  continuation